HLMamba: Hybrid Lightweight Mamba-Based Fusion Network for Dense Prediction of Remote Sensing Images

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***Abstract*— Convolutional neural network (CNN)- and transformer-based methods have made significant advances in dense prediction tasks for remote sensing images (RSIs). Nevertheless, both architectures have inherent limitations: CNNs are constrained by a finite receptive field, rendering capturing long-range dependencies challenging, whereas transformers face challenges in terms of ensuring computational efficiency. Recently, the mamba architecture, based on state-space models (SSMs), has emerged as a promising alternative by enabling efficient modeling of long-term dependencies while reducing linear computational complexity. In this study, we present HLMamba, a hybrid lightweight mamba fusion network that integrates a CNN-based encoder with mamba-based fusion modules and a mamba-based decoder. Specifically, after feature extraction using a CNN-based encoder, we introduce a comprehensive interaction mamba (CIM) to enhance modality fusion by capturing intramodal feature dependencies and facilitating intermodal feature interactions, resulting in more comprehensive fusion features. Moreover, we design a wavelet complementary hybrid mamba (WCHM) to integrate fused features by exploring their correlations in feature and wavelet space for accurate dense prediction. Experimental results on two widely used datasets, ISPRS Vaihingen and ISPRS Potsdam, demonstrate the effectiveness and potential of the proposed HLMamba. By leveraging linear complexity and global modeling capabilities, HLMamba achieves better efficiency and accuracy in RSI analysis than existing transformer-based and CNN-based models. The source code is available at: https://github.com/MAXHAN22/HLMamba**

***Index Terms*— Dense prediction, mamba, remote sensing images (RSIs), wavelet.**

I. Introduction

DENSE prediction of remote-sensing images (RSIs) is a pivotal task in this field, with the aim of annotating each pixel of an image according to its category. Currently, dense prediction tasks are predominantly implemented using deep learning approaches [1], [2], which can be classified into two fundamental types: convolutional neural network (CNN)- [3], [4], [5], [6], [7], [8] and transformer-based methods [9], [10], [11]. CNN-based approaches progressively extract features from shallow to deep layers by stacking convolutional operations on the image data. Conversely, transformer-based methods utilize self-attention mechanisms to capture global dependencies flexibly.

Notably, RSIs have considerably high spatial resolutions, exhibiting large spatial-scale variations within and between targets [12]. Hence, although CNN- and transformer-based methods offer distinct advantages, they have inherent constraints when applied to RSIs. In CNN-based methods, the inherent limitation of a small receptive field can lead to the loss of critical feature representations that are essential for capturing local features. Although this flaw can be addressed by enlarging the receptive field using dilated convolutions [13] or enhancing feature recognition through residual connections [14], these approaches often result in a significant increase in network computation, rendering them suboptimal solutions. In contrast, transformer-based methods are valued for their ability to efficiently capture global contextual information and model spatial dependencies through self-attention mechanisms [15]. However, because of the preprocessing step of dividing images into smaller patches, these methods must compute attention across all patches, resulting in a computational complexity that increases exponentially with resolution. Thus, developing more efficient architectures that can be fully adapted to RSIs is crucial.

The application of deep learning to multimodal fusion has advanced significantly. This technique has proven to be an effective image enhancement method, improving the ability of the model to handle complex scenes [16], [17]. Typically, these models employ the following three strategies. The first strategy involves directly merging features through simple element-level operations, such as concatenation, addition, multiplication, or their combinations [18], [19], [20]. Although this approach is simple and efficient, it fails to address intermodal distinctions. The second strategy incorporates specific techniques, such as combining various convolution and pooling operations [21], [22], directly using modules from various transformers [23], and even utilizing frequency-domain information to aid fusion [24]. Although these methods facilitate feature interactions, they also have

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the aforementioned limitations. The third strategy introduces various attention [25], [26], [27] and gating mechanisms [28] into the fusion process, mainly by weighting the information to reduce redundancy. However, these methods increase the computational burden and tend to focus predominantly on intermodal feature interactions, ignoring the critical role of global and intramodal information mining in dense prediction.

The emergence of the mamba architecture presents promising solutions to these challenges, particularly because of its ability to capture global contextual information using embedded state-space models (SSMs). SSMs incorporate time-varying parameters that enable efficient training and representation of the mamba state-space equations in a recurrent form, exhibiting linear complexity. When combined with a ubiquitous multilayer perceptron, this forms the most basic mamba structure [29]. Notably, the incorporation of a scanning mechanism [30] equips mamba with the ability to select pivotal information for modeling long sequences while simultaneously capturing long-term dependencies. Its exceptional performance in modeling the global context suggests that the mamba structure can be highly effective in RSI-based fusion networks. Nonetheless, a few potential problems remain when mamba is applied to RSI tasks. First, recent models tend to over-rely on mamba for modeling global dependencies, often at the expense of effectively modeling local features. Second, the original mamba framework treats each token equally in the spatial domain, focusing only on the feature space, which results in the loss of critical information regarding the pixels located at the boundaries.

To address the aforementioned challenges, this article presents HLMamba, a novel hybrid lightweight mamba-based fusion network, designed to effectively mine both intra- and inter-modal information while aggregating multiscale information from two distinct feature spaces. Specifically, we propose a comprehensive interaction mamba (CIM) that encourages complementary intermodal feature learning after mining intramodal dependencies. Additionally, we propose a wavelet complementary hybrid mamba (WCHM), which employs a local spatial capture unit (LSCU) and 2-D vision SSM (2DSSM), to aggregate multiscale features across local and global domains. Subsequently, we devise a wavelet allocation module (WAM) to incorporate wavelet features that provide additional useful information. Overall, HLMamba captures both the global and local bi-domain dependencies of fused features, enabling the application of smaller parameters while delivering a strong performance on RSIs.

The main contributions of the study are summarized as follows.

1) We propose HLMamba, a novel fusion network that incorporates mamba for the dense prediction of RSIs. HLMamba addresses the limitations in the mamba structure by overcoming those related to direct feature fusion and ensuring that critical information is not ignored during aggregation. This approach fully leverages the potential of mamba for multimodal RSIs.
2) The proposed CIM exploits the characteristics of the mamba structure to achieve a more comprehensive intramodal feature representation while promoting intermodal feature interactions. We design a WCHM to combine the global features extracted by the 2DSSM with the local features captured by the LSCU, utilizing parallel WAM to assign wavelet information and achieve multiscale feature aggregation in a wavelet-feature space.
3) Extensive experiments on two publicly available RSI datasets, i.e., ISPRS Potsdam and ISPRS Vaihingen, demonstrate that HLMamba outperforms CNN-, transformer-, and mamba-based methods in the dense prediction of RSIs while maintaining low computational complexity.

II. Related Work

A. Dense Prediction of RSIs

Dense prediction of RSIs has advanced significantly, with a plethora of diverse deep learning approaches that are none more glamorous than CNN- and transformer-based methods.

CNN-based models are capable of spontaneously learning spatial hierarchical features from images, making them well-suited for dense prediction tasks in RSIs owing to their strong ability to capture local features. Li et al. [32] developed a lightweight SeaNet with separate matching modules designed to handle low- and high-level information independently according to their characteristics. Wang et al. [33] replaced standard convolution with depth-wise dilated convolution, utilizing fewer parameters to effectively capture the multiscale variations of spatial features in RSIs. To overcome the limitations of the inherently small receptive field and cope with scale variations, several models have introduced multiscale modules. Xu et al. [5] designed HRCNet, which uses a feature pyramid structure to retain spatial information across different scales. Zhong et al. [8] take full advantage of the capability of CNNs to perceive local spatial structures by employing a proposed multilevel convolutional module that decodes multiscale information step by step. Conversely, Yang et al. [34] introduced an attention mechanism to make the model applicable to RSIs by adaptively selecting the features across multiple branches.

Unlike traditional CNNs, which struggle to extract sufficient global spatial features and dependencies from RSIs, the transformer, a deep learning model based on the self-attention mechanism, successfully overcomes this challenge. Wang et al. [35] employed the Swin transformer [36] to extract the backbone of contextual information and efficiently aggregate multiscale features through dense connectivity. Zhang et al. [22] used low-level features rich in detail information as cues to guide the aggregation of high-level features. Recent models have also paid sufficient attention to the importance of local context, combining both for dense predictions. Sha and Li [37] extended an attention-based multilayer perception module at the end of each level to enhance local feature mining. Zhang et al. [38] directly proposed a hybrid model that uses the CNN component to extract local features and the transformer component to extract global information.

Transformer-based and hybrid models cannot be deployed on real-world devices owing to the secondary complexity of

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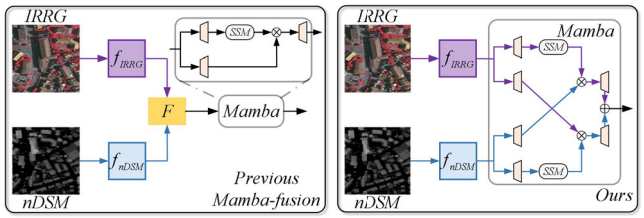


Fig. 1. Comparison of previous methods with CIM of our HLMamba. (a) Previous methods. (b) Our HLMamba.

self-attention mechanisms. The emergence of mamba-based models addresses this immediate need.

B. Mamba-Based Methods in RSIs

Researchers are currently exploring the application of mamba-based models in RSIs. Guo et al. [39] successfully modeled global information using mamba with only linear complexity. Zhao et al. [40] introduced for the first time the mamba model in the RSI tasks. Ma et al. [41] incorporated SSM-based auxiliary branches into the CNN-based backbone to provide additional global information for predictions. Chen et al. [42] improved the underlying SSM to enhance mamba modeling using the proposed multipath activation mechanism. Researchers have explored the application of mamba during the fusion phase. Xie et al. [43] first used pooling and simple element-wise operations to obtain fused features before feeding them into a mamba block to mine the information correlation between different modalities. Li et al. [44] and He et al. [45] performed channel swapping operations on multimodal features, which were then fed into the mamba to obtain cross-modal correlations after integrating information from different channels. In summary, as illustrated in Fig. 1(a), these methods first perform feature fusion and then feed the results into mamba to explore cross-modal dependencies, as mamba itself lacks a branch structure similar to that of cross-attention [46].

We argue that mamba has not yet fully realized its potential in RSI tasks. Unlike existing works, first, we can explore the fusion of multimodal features directly within the internal structure of mamba, making full use of the structure. Second, during the fusion process, in addition to focusing on cross-modal feature dependencies, our model focuses on to the relationships between intra-modal features. Moreover, it is not obvious whether the model is prone to losing information regarding pixels located at the boundaries during the prediction tasks of RSIs compared to normal images, and focusing solely on the feature space may lead to the loss of some important information [47].

Inspired by the above observations, we designed the CIM and WCHM for HLMamba. As shown in Fig. 1(b), our approach differs from other approaches in that it works directly within mamba to encourage inter-modal interactions while exploring intra-modal feature dependencies. Moreover, we propose WCHM, which facilitates the aggregation of global and local multiscale features from differentiated feature spaces and wavelet spaces.

C. State-Space Model (SSM)

SSMs are central to establishing long-term dependencies in mamba, originating as a mathematical model in control theory describing the behavior of dynamical systems. Initially, SSMs did not receive much attention from researchers because of their parametric and computational intensity. The structured state-space sequential model (S4) [48] employed a normal plus low-rank approach that utilized a normalized parameterization tactic, signaling the availability of SSMs for use in neural network architectures. Smith et al. [49] built on the design of S4 and introduced a new state-space layer that constitutes a multi-input, multi-output SSM, referred to as S5.

To increase the potential of SSMs, the recent advancements in mamba have introduced time-varying parameters and set up specific perceptual algorithms for effective training, effectively enabling the modeling of long sequences, which has gradually become prevalent in visual tasks. Chen et al. [50] proposed performing before and after selective scanning in specific directions, and Zhu et al. [51] implemented selective scanning in the horizontal direction; however, both approaches faced the problem of orientation sensitivity. Most notably, VMamba [31] included a novel cross-scanning mechanism called 2D-selective-scan (SS2D) to solve the above dilemma. In contrast to other scanning mechanisms, SS2D, which is widely used in [52] and [53], employs scanning from four vertices to their opposite positions to traverse the entire image, ensuring that each pixel carries information from all other positions, modeling global information while requiring only a linear amount of computation.

Based on the above description, we adopted SS2D as the scan mechanism of HLMamba to effectively model the feature dependencies in CIM and WCHM.

III. Proposed Approach

In this section, we present a brief overview of HLMamba's holistic design. We introduce CIM, which leverages the strengths of the 2DSSM to fully exploit intramodal information and promote cross-modal feature interactions. Finally, we discuss the WCHM, which consists of a global component based on mamba and a local component based on convolutional layers, incorporating WAM to provide crucial wavelet information for decoding.

A. Network Architecture

The overarching architecture of HLMamba, structured as an encoder–decoder [54], is shown in Fig. 2. To balance between extraction capability and parameter efficiency, we selected MobileNet V2 [55] as the encoder to extract feature information from infrared–red–green (IRRG) and normalized digital surface model (nDSM) images in parallel, denoted as R_i and D_i ($i = \{1, 2, 3, 4, 5\}$). Features R_i and D_i are sequentially fed into the corresponding CIM modules at each level, where the multimodal features undergo intra- and intermodal interactions to produce features F_i. Finally, a multilevel decoder, which includes a patch-embedding operation and two WCHMs, is used to process F_i through concatenation to generate the final prediction map. In this architecture, the 2DSSM structure aggregates multiscale features in the global domain, while

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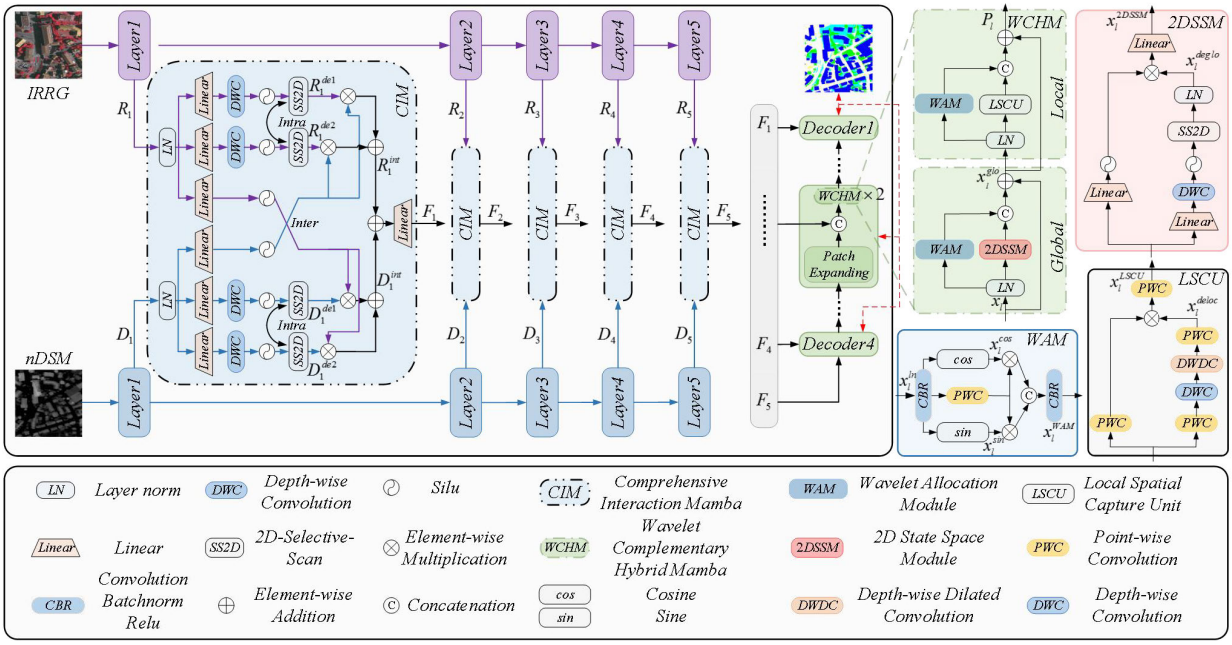


Fig. 2. Overview of our HLMamba overall framework.

the CNN-based LSCU aggregates them in the local domain, with WAM providing the necessary wavelet information for decoding.

B. Comprehensive Interaction Mamba (CIM)

Multimodal data can exhibit dependencies on both local and global features, and mamba-based models have demonstrated outstanding potential for modeling these correlations. Nevertheless, current mamba architectures [43], [44], [45] are not inherently equipped to directly process multimodal information. Hence, we designed a CIM module that fully exploits intramodal dependencies while enabling effective and sufficient intermodal information interactions, resulting in a more comprehensive multimodal representation.

Considering R_1 and D_1 as examples, the inputs are first normalized using the layer-norm operation and then passed through a linear layer to form three branches. Two of these branches proceed through depth-wise convolution (DWC), sigmoid linear unit (SiLU) activation function, and SS2D to capture two distinct long-term dependencies R_1^{de1}, R_1^{de2} and D_1^{de1}, D_1^{de2} in their respective modalities, denoted as follows:

$$\begin{cases} R_1^{\mathrm{de1}}, \quad R_1^{\mathrm{de2}} = \mathrm{SS2D}(\mathrm{Silu}(\mathrm{DWC}(\mathrm{Linear}(\mathrm{LN}(R_1))))) \\ D_1^{\mathrm{de1}}, \quad D_1^{\mathrm{de2}} = \mathrm{SS2D}(\mathrm{Silu}(\mathrm{DWC}(\mathrm{Linear}(\mathrm{LN}(D_1)))))) \end{cases} \tag{1}$$

where LN, Linear(·), Silu(·), and DWC(·) represent the layer norm operation, linear layer, SiLU activation, and DWC, respectively. To facilitate the intermodal feature interaction, the remaining branches are modulated using a gating factor and added to the other modality to obtain the intermodal features R_1^{inter}, D_1^{inter}. Finally, R_1^{inter} and D_1^{inter} are combined to form a fused representation F_1, which can be expressed as follows:

$$\begin{cases} R_1^{\mathrm{inter}} = \mathrm{Linear}(\mathrm{Silu}(\mathrm{LN}(D_1))) \times \left(R_1^{\mathrm{de1}} + R_1^{\mathrm{de2}}\right) \\ D_1^{\mathrm{inter}} = \mathrm{Linear}(\mathrm{Silu}(\mathrm{LN}(R_1))) \times \left(D_1^{\mathrm{de1}} + D_1^{\mathrm{de2}}\right) \end{cases} \tag{2}$$

$$F_1 = \mathrm{Linear}\left(R_1^{\mathrm{inter}} + D_1^{\mathrm{inter}}\right) \tag{3}$$

where "×" and "+" indicate element-wise multiplication and addition, respectively.

C. Wavelet Complementary Hybrid Mamba (WCHM)

The dense prediction of RSIs has benefited considerably from models integrating global and local information, and the recent increase of mamba-based methods has introduced new concepts in this field. Therefore, we propose a WCHM to explore global and local representations in the wavelet-space dual domain.

Specifically, WCHM processes multiscale features $x_l (l = \{1, 2, 3, 4\})$ through two parallel branches to aggregate information x_l^{glo} with global representations. After a layer-norm step, the normalized feature is sent to the 2DSSM in one branch to capture long-term spatial information, while in the other branch, WAM is used to provide additional wavelet cues for decoding. These two domains of information are then aggregated and combined with the original input through a residual connection, which can be formulated as follows:

$$x_l^{\mathrm{glo}} = \mathrm{Cat}(\mathrm{2DSSM}(\mathrm{LN}(x_l))), \quad \mathrm{WAM}(\mathrm{LN}(x_l))) + x_l \tag{4}$$

where 2DSSM(·), WAM(·), and Cat(·) represent 2DSSM, WAM, and element-wise concatenation, respectively. Subsequently, x_l^{glo} is modeled to extract local features. Following a similar process, we also normalize x_l^{glo} with the difference of utilizing LSCU to capture spatial locality. The WAM continues to provide wavelet cues. Moreover, a residual operation is applied to generate the prediction map P_l enriched with local information. The overall operation can be expressed as follows:

$$P_l = \mathrm{Cat}\left(\mathrm{LSCU}\left(\mathrm{LN}\left(x_l^{\mathrm{glo}}\right)\right)\right), \quad \mathrm{WAM}\left(\mathrm{LN}\left(x_l^{\mathrm{glo}}\right)\right)\right) + x_l^{\mathrm{glo}} \tag{5}$$

where LSCU(·) denotes the LSCU module.

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D. 2DSSM

To effectively mine global information while maintaining model efficiency, we introduce a 2DSSM with linear complexity into our WCHM module. Similar to the dependency acquisition operation in CIM, the feature x_l is processed in two parallel branches following layer norm. In the first branch, x_l sequentially passes through a linear layer, DWC, SiLU activation, SS2D, and another layer norm to obtain x_l^{deglo}. In the second branch, x_l is processed only through the linear layer and SiLU activation. The outputs of both branches are then combined to form the final output x_l^{2DSSM}, which can be expressed as follows:

$$x_l^{\text{deglo}} = \text{LN}(\text{SS2D}(\text{Silu}(\text{DWC}(\text{Linear}(\text{LN}(x_l)))))) \tag{6}$$

$$x_l^{\text{2DSSM}} = \text{Linear}\left(\text{Silu}\left(\text{Linear}(\text{LN}(x_l)) \times x_l^{\text{deglo}}\right)\right). \tag{7}$$

E. Local Spatial Capture Unit (LSCU)

After obtaining global information using the 2DSSM, we assign relevant locality to improve its predictive properties. Recent models have often employed multilayer perceptions for this function. However, this approach can significantly increase the training burden owing to the wide channels of the perceptions. Drawing inspiration from a single gate [56] and the 2DSSM architecture, we designed a lightweight LSCU (Fig. 2) that effectively captured stable local contextual information by selectively augmenting local features across the spatial domain.

Specifically, we emulate the two-branch architecture of the standard 2DSSM. In one branch, we reduce the number of parameters by decomposing a standard convolution operation into a combination of depth-wise, depth-wise dilated, and point-wise convolutions to produce x_l^{deloc}. Depth-wise and depth-wise dilated convolutions efficiently capture the long-range relationships of local features, whereas point-wise convolution is used to restore dimensionality. In contrast to the 2DSSM, which applies a SiLU activation in the other branch to multiply features with established long-range relationships, inspired by the efficiency of single gates, we directly weigh the feature map to capture local information more efficiently. The entire process is expressed as follows:

$$x_l^{\text{deloc}} = \text{PWC}\left(\text{DWDC}\left(\text{DWC}\left(\text{PWC}\left(\left(\text{LN}\left(x_l^{\text{gLo}}\right)\right)\right)\right)\right)\right) \tag{8}$$

$$x_l^{\text{LSCU}} = \text{PWC}\left(x_l^{\text{deloc}} \times \text{PWC}\left(\text{LN}\left(x_l^{\text{gLo}}\right)\right)\right) \tag{9}$$

where DWDC$(\cdot)$ and PWC$(\cdot)$ denote depth-wise dilated and point-wise convolutions, respectively.

F. Wavelet Allocation Module (WAM)

To achieve spatial-wavelet dual-domain representation at both global and local levels, we embedded the WAM with the 2DSSM and LSCU in a head-to-tail configuration, as displayed in Fig. 2. Wavelets can comprehensively represent features [57] and have been widely adopted in several areas of deep learning [58], [59] owing to their capacity to generate rich contextual cues while adapting to complex scenes across various scales.

In contrast to directly applying the wavelet transform [60], which is computationally intensive, we used cosine and sine functions, the simplest forms of wavelets [57], to provide additional wavelet information for decoding. As shown in Fig. 2, the input features x_l^{ln} are first processed through a 1×1 convolution block and subsequently passed through sine and cosine functions. As mentioned earlier, inspired by the efficiency of single gates [56], we applied a single-layer DWC to weigh x_l^{sin} and x_l^{cos}, replacing the traditional attention mechanism with simpler operations such as pooling combined with activation to form the attention graph x_l^{WAM}. The key process of WAM is expressed as follows:

$$\begin{cases} x_l^{\text{sin}} = \text{Sin}\left(\text{CBR}\left(x_l^{\text{ln}}\right)\right) \\ x_l^{\text{cos}} = \text{Cos}\left(\text{CBR}\left(x_l^{\text{ln}}\right)\right) \end{cases} \tag{10}$$

$$x_l^{\text{WAM}} = \text{Cat}\left(x_l^{\text{sin}} \times \text{PWC}(\text{CBR}(x_{\text{ln}})), x_l^{\text{cos}} \times \text{PWC}(\text{CBR}(x_{\text{ln}}))\right). \tag{11}$$

G. Loss Function

To generate accurate dense predictions, labels were used to supervise the outputs at all four levels of the decoders, ensuring that each decoder in the model contributes effectively to the final result. The total loss is expressed as follows:

$$L_{\text{sup}} = \sum_{l=1}^{4} L_{\text{ce}}(P_l, \text{Label}), \quad l = \{1, 2, 3, 4\} \tag{12}$$

$$L_{\text{ce}} = -\sum_{n}^{6} Y_n \times \log(X_n), \quad n = \{0, 1, 2, 3, 4, 5\} \tag{13}$$

where $L_{\text{ce}}(\cdot)$ represents the cross-entropy loss. Here, Y_i is an indicator variable that equals one if the predicted category matches the labeled category, and zero otherwise. X_n denotes the class probability distribution of all the pixel samples.

IV. Experimental Settings and Results

To verify the efficiency and superiority of HLMamba in dense prediction tasks, we experimented with two public datasets: ISPRS Vaihingen and ISPRS Potsdam. In this section, we describe the training setup and implementation details. In addition, ablation studies were performed to demonstrate the effectiveness of the proposed modules. In particular, we performed tests on the RGBD-Mirror and LoveDA datasets to verify the generalization ability of HLMamba. Finally, we studied the performance of our model in extreme environments.

A. Datasets

1) *Potsdam:* This dataset [61] contains 24 pairs of images with a spatial resolution of 5 cm, encompassing six categories: impervious surfaces, trees, low vegetation, buildings, cars, and clusters. Each image features a particularly high resolution with an average size of 6000×6000 pixels. For this study, 17 pairs were used as

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TABLE I
PERFORMANCE COMPARISON BASED ON VAIHINGEN. THE BEST RESULTS IN EACH COLUMN ARE HIGHLIGHTED IN RED, WHILE THE SECOND-BEST IN BLUE

	Imp.surf		Building		Low veg.		Tree		Car		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
U-Net$_{2015}$[52]	91.68	80.90	89.84	86.50	77.97	65.91	91.30	77.86	75.80	71.22	79.75	71.34
FCN-Net$_{2017}$[3]	89.66	79.71	93.22	86.80	75.83	64.33	89.22	75.58	45.12	40.16	78.61	69.32
SegNet$_{2017}$[4]	89.88	80.93	90.88	86.54	78.66	64.07	88.96	75.96	43.93	43.16	78.46	70.13
HRCNet$_{2021}$[5]	91.62	81.60	91.72	88.01	79.24	67.38	90.55	78.58	70.69	68.73	84.76	76.86
MANet$_{2022}$[6]	91.57	83.23	93.77	90.04	76.29	65.59	91.89	77.94	86.03	81.01	87.91	79.56
EAEFNet$_{2023}$[25]	93.78	84.37	95.35	92.25	79.87	67.98	89.11	79.00	81.54	74.24	87.93	79.57
CEKD$_{2023}$[7]	92.22	85.05	95.78	91.66	76.57	66.77	92.69	78.97	76.63	72.01	86.71	78.89
MGCNet$_{2022}$[28]	91.14	78.58	89.92	85.50	72.25	61.33	90.50	74.58	18.60	18.43	72.48	63.68
SeaNet$_{2023}$[32]	91.18	84.89	95.58	91.52	76.22	65.74	92.76	78.53	79.56	75.33	87.06	79.20
CAINet$_{2024}$[21]	90.50	83.10	95.16	91.22	77.28	64.94	91.16	77.76	54.11	51.07	81.64	73.62
MAGNet$_{2024}$[8]	92.16	83.09	94.89	91.56	84.16	65.87	82.85	75.63	66.45	63.17	84.10	75.87
CMX$_{2022}$[22]	91.78	85.39	95.15	91.28	77.00	66.29	92.59	78.85	83.23	78.58	87.95	80.08
FTUNetFormer$_{2022}$[10]	89.42	78.02	89.34	83.88	70.78	60.15	91.12	74.09	53.42	50.73	78.82	69.38
CATNet$_{2024}$[26]	90.53	83.98	95.27	91.08	78.27	65.62	90.57	77.72	70.28	61.85	84.98	76.05
CPNet$_{2024}$[27]	91.79	78.58	93.99	88.97	61.96	55.58	90.77	74.31	29.63	27.68	74.23	65.02
MedMamba$_{2024}$[50]	88.28	75.36	90.90	83.42	70.00	58.15	86.47	70.74	10.12	9.75	69.15	59.48
Ours	91.18	85.08	96.15	93.04	80.46	67.88	90.91	79.16	87.42	81.20	89.22	81.27

the training set, while the remaining seven pairs served as the test set.

2) *Vaihingen:* This dataset [61] comprises 16 pairs of IRRS and nDSM images, each with a spatial resolution of 9 cm and a similarly high average image resolution of 2500 × 2500 pixels. It includes five foreground classes: impervious surfaces, trees, low-vegetation areas, buildings, and cars. In this study, the dataset was divided into a training set containing 12 pairs and test set containing four pairs.

B. Training Settings

1) Data Augmentation: To demonstrate the effectiveness of the proposed method, we used direct data enhancement techniques. Because of the large size of the images, we followed the approach of Peng et al. [62], resizing the input images to 256 × 256 pixels. To further enhance the dataset, we applied various enhancement techniques including scaling, sliding windows, random cropping, horizontal and vertical flipping, and counterclockwise rotations of 90°, 180°, and 270° to all inputs. Consequently, more than 30 000 images were obtained for the training set and approximately 4000 images for the test set from the Potsdam dataset, while approximately 17 000 images were obtained for the training set and more than 400 images for the test set from the Vaihingen dataset.

2) Training Details: The experiments on the two datasets were conducted using a uniform training setup. The proposed method was compared to several established methods to benchmark its performance. All experiments were performed using the PyTorch framework on a single PC equipped with 12 GB RAM and an NVIDIA TITAN V GPU. The Adam optimizer [63] was employed for training, with an initial learning rate of 0.0001, a decay coefficient of 0.0005, and a batch size of 12. The learning rate was decreased to one–tenth of the original value every 20 epochs, and the model was trained for 100 epochs.

3) Evaluation Metrics: We employed four key evaluation metrics to evaluate the performance of the proposed model: intersection over union (IoU), class accuracy (Acc), mean IoU (mIoU), and mean Acc (mAcc) [64], considering both the individual class and overall perspectives.

C. Performance Comparison

1) Benchmark Models: To verify the superiority of the proposed models, we selected a series of highly competitive state-of-the-art (SOTA) models as benchmarks. CNN-based methods included U-Net [54], FCN-Net [3], SegNet [4], HRCNet [5], MANet [6], EAEFNet [25], CEKD [7], MGCNet [28], SeaNet [32], CAINet [21], and MAGNet [8]. Transformer-based models included CMX [22], FTUNetFormer [10], CATNet [26], and CPNet [27]. Additionally, MedMamba [50] was included as a mamba-based method that employs SS2D as the scanning mechanism.

2) Quantitative Perspective: Tables I and II present the quantitative performance of HLMamba on the ISPRS Vaihingen and Potsdam datasets. HLMamba achieved the highest or second-highest scores across nearly all metrics, demonstrating its impressive dense prediction performance on various RSI datasets. For the Vaihingen dataset, HLMamba excelled, particularly in predicting buildings, low vegetation, and car categories. For the Potsdam dataset, the algorithm predicted categories of buildings, low vegetation, and clusters. Notably, while SS2D is highly effective in capturing nonlocal dependencies, MedMamba underperforms on complex RSIs, demonstrating that the straightforward application of mamba is insufficient for RSI tasks and that it requires further enhancement. Similarly, these results also indicate that our HLMamba concept is effective for fusing and aggregating multimodal features.

3) Qualitative Perspective: Dense prediction visualizations of the ISPRS Vaihingen and Potsdam datasets are shown in Fig. 3, featuring four groups from Vaihingen and three groups from Potsdam. These visual comparisons clearly demonstrate the effectiveness of HLMamba in RSI tasks. Specifically, when addressing complex scenes with multiple categories and scales, as illustrated in rows three, four, and five, HLMamba outperforms other methods in accurately predicting cars (yellow) and crowds (red) with high clarity. This highlights the

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TABLE II
PERFORMANCE COMPARISON BASED ON POTSDAM

	Imp.surf		Building		Low veg.		Tree		Car		Cluter		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
U-Net$_{2015}$[52]	90.03	80.27	88.71	82.92	85.82	71.60	84.06	72.05	93.89	90.24	50.30	36.26	82.13	72.22
FCN-Net$_{2017}$[3]	89.47	79.77	90.69	83.60	85.13	71.12	82.86	71.23	91.02	81.53	49.05	36.49	78.61	69.32
SegNet$_{2017}$[4]	90.18	80.46	90.21	84.18	85.88	71.63	82.49	70.68	93.15	89.72	51.76	37.21	82.28	72.31
HRCNet$_{2021}$[5]	90.03	81.68	90.87	85.75	88.17	73.18	82.02	71.32	93.94	89.82	56.72	40.03	83.63	73.63
MANet$_{2022}$[6]	90.48	81.20	87.58	83.47	88.78	72.74	82.69	71.54	90.96	88.73	46.16	38.11	81.11	72.63
EAEFNet$_{2023}$[25]	92.32	82.04	93.22	89.54	84.34	72.37	85.02	71.91	94.64	88.30	53.55	39.34	83.85	73.92
CEKD$_{2023}$[7]	91.09	80.89	93.37	88.97	84.70	71.71	85.18	71.62	93.01	89.35	52.63	38.18	83.33	73.45
MGCNet$_{2022}$[28]	89.80	78.77	88.34	81.99	86.87	69.48	74.97	63.17	90.37	86.52	42.13	34.30	78.75	69.04
SeaNet$_{2023}$[32]	90.61	81.21	93.87	89.02	86.72	72.67	83.70	72.51	93.39	88.91	53.24	39.66	83.59	74.00
CAINet$_{2024}$[21]	90.52	80.35	93.13	89.32	84.70	70.55	81.15	67.89	93.48	85.33	48.06	37.04	81.84	71.75
MAGNet$_{2024}$[8]	89.51	80.26	92.76	89.13	83.20	70.51	83.19	65.17	94.08	83.16	51.76	33.98	82.42	70.37
CMX$_{2022}$[22]	90.54	81.86	93.80	89.28	87.17	73.11	83.52	71.45	92.23	87.99	51.61	40.58	83.14	74.04
FTUNetFormer$_{2022}$[10]	87.77	77.93	87.98	81.38	86.68	68.23	72.06	59.72	92.40	87.52	44.98	35.54	78.65	68.39
CATNet$_{2024}$[26]	88.12	79.80	93.32	89.52	85.91	70.66	78.78	65.85	91.56	84.99	55.58	40.41	82.21	71.87
CPNet$_{2024}$[27]	89.36	75.08	87.59	85.02	85.99	67.01	73.81	60.45	39.93	38.42	36.54	25.91	68.87	58.65
MedMamba$_{2024}$[50]	90.87	74.56	87.31	79.69	84.80	66.71	70.16	60.13	82.19	73.32	29.46	26.37	74.13	63.46
Ours	91.01	83.03	93.52	89.73	86.23	73.25	85.30	73.19	94.03	89.50	58.09	42.73	84.70	75.24

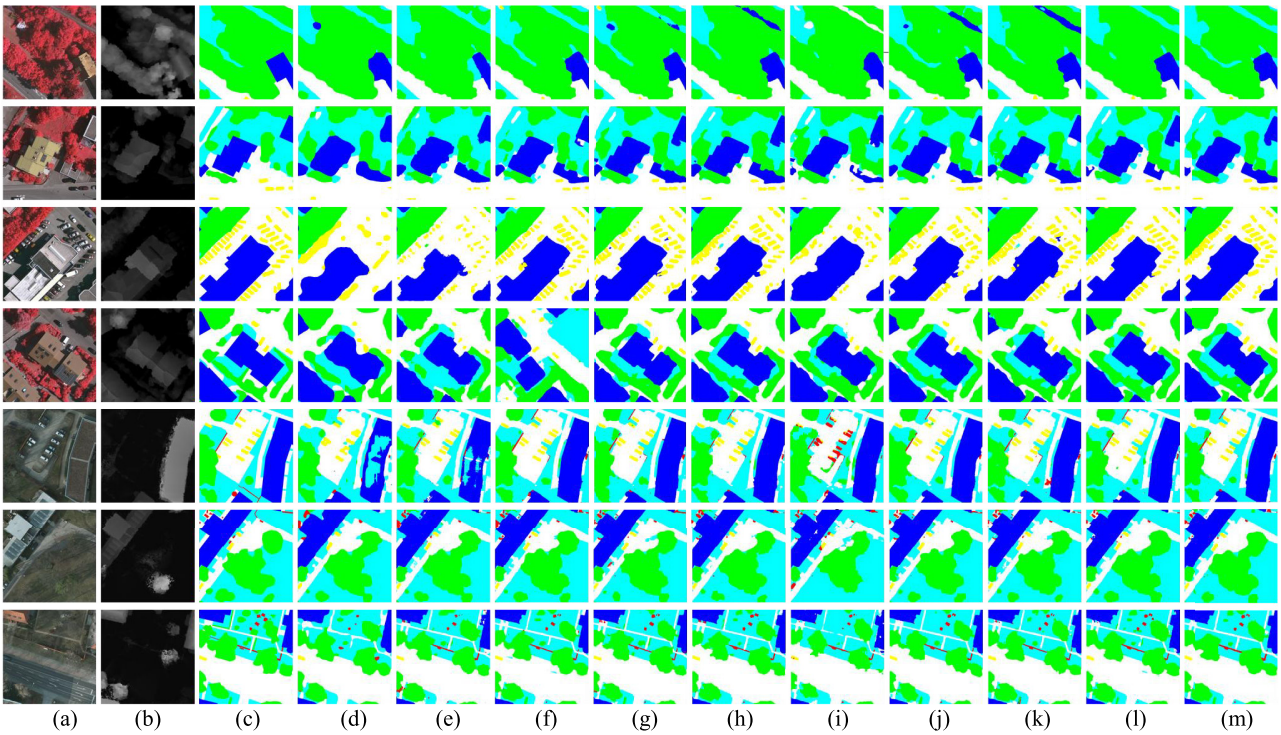


Fig. 3. Dense prediction performance (a) IRRG, (b) nDSM, (c) Label, (d) FCN-Net, (e) SegNet, (f) EAEFNet, (g) CEKD, (h) CMX, (i) CPNet, (j) CATNet, (k) MAGNet, (l) MedMamba, and (m) HLMamba (Ours).

efficacy of the CIM module in directly and effectively using multimodal information. Furthermore, rows two, five, and seven illustrate that our model generates accurate distributions of detailed information, such as buildings, low vegetation, crowds, and boundaries of vegetation, outperforming the other models. These visual comparisons further demonstrate the ability of HLMamba to capture fine details because while 2DSSM and LSCU fully aggregate global–local features, WAM provides essential wavelet information for accurate predictions.

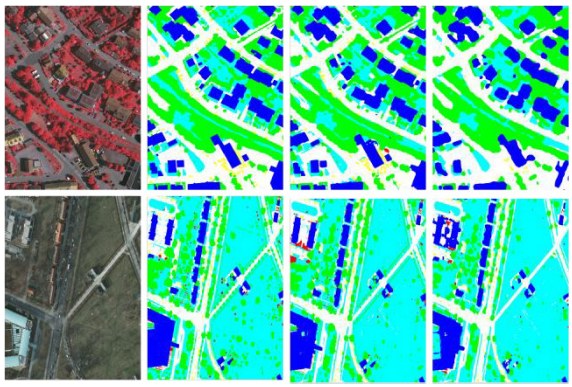
To analyze the results from a global perspective, we also provide two sets of complete original-scale remote sensing dense predictions without the mentioned resizing operations (Fig. 4). These figures clearly and intuitively indicate that our proposed HLMamba achieves outstanding dense prediction performance.

4) Complexity Comparison: We compared the performance of HLMamba with that of other SOTA models in terms of number of parameters (Params) and floating-point operations (Flops) to evaluate the size and computing speed, respectively (Table III). HLMamba, which implements SS2D for modeling long-range dependencies with linear computational complexity, requires only 8.40 M Params and 2.92 G Flops. This is significantly lower than that required by other CNN-, transformer-, and mamba-based models, with the exception of SeaNet, which was designed for unimodal modalities. Nevertheless, the performance of HLMamba on the two datasets was significantly superior to that of SeaNet.

D. Ablation Studies

In this section, we explore the contribution of the main HLMamba modules, including CIM, WCHW, and their

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(a) (b) (c) (d)

Fig. 4. Original-scale dense prediction performance (a) IRRG, (b) Label, (c) HLMamba (Ours), and (d) MedMamba.

TABLE III
COMPLEXITY COMPARISON RESULTS

	Type	Backbone	Params(M)	Flops(G)
U-Net$_{2015}$[52]	Uni	VGG-16	26.36	55.93
FCN-Net$_{2017}$[3]	Uni	VGG-16	134.29	74.55
SegNet$_{2017}$[4]	Uni	VGG-16	53.56	18.3
HRCNet$_{2021}$[5]	Uni	ResNet-50	62.71	30.27
MANet$_{2022}$[6]	Uni	ResNet-50	35.86	19.49
EAEFNet$_{2023}$[25]	Dual	ResNet-50	77.97	23.3
CEKD$_{2023}$[7]	Uni	ResNet-50	245.03	68.61
MGCNet$_{2022}$[28]	Dual	ConvNeXt	209.99	47.51
SeaNet$_{2023}$[32]	Uni	Mobile-Net v2	2.76	1.33
CAINet$_{2024}$[21]	Dual	Mobile-Net v2	13.8	14.67
MAGNet$_{2024}$[8]	Dual	Mobile-Net v2	16.18	9.90
CMX$_{2022}$[22]	Dual	SegFormer	66.56	14.26
FTUNetFormer$_{2022}$[10]	Uni	Swin-Transformer	36.17	16.85
CATNet$_{2024}$[26]	Dual	Swin-Transformer	262.74	76.79
CPNet$_{2024}$[27]	Dual	Swin-Transformer	222.91	57.56
MedMamba$_{2024}$[50]	Uni	mamba	23.50	35.12
Ours	Dual	Mobile-Net v2	8.40	2.92

TABLE IV
ABLATION EXPERIMENTS

Baseline	CIM	WCHM	Vaihingen		Potsdam	
			mAcc	mIoU	mAcc	mIoU
√			84.05	76.29	83.29	73.43
√	√		86.29	78.44	83.73	73.76
√		√	87.49	79.44	84.18	73.85
√	√	√	**89.22**	**81.27**	**84.70**	**75.24**

variants. We used the Vaihingen and Potsdam datasets for the ablation experiments, following the same setup. The results are presented in Tables IV–VI, where the baseline was obtained by excluding all the designed modules and replacing CIM and WCHM with element-wise addition operations carrying standard convolutional blocks and four two-layer residual blocks, respectively.

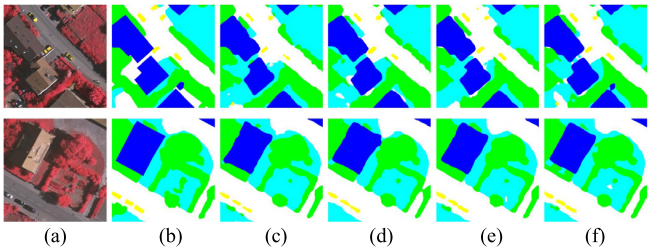
1) Contribution of the CIM: The results in Table IV show that the CIM effectively implements feature fusion within the mamba structure. Adding the CIM module to the baseline improved the mAcc and mIoU by 2.24% and 2.15% on the Vaihingen dataset, respectively, and by 0.44% and 0.33% on the Potsdam dataset, respectively. Fig. 5 demonstrates that the SRF module effectively fuses multimodal information, thereby improving segmentation performance in complex scenes. The figure also shows the predictions of the baseline and the proposed modules. The addition of the CIM enables the model to address complex scenarios.

TABLE V
ABLATION EXPERIMENTS WITH WCHM

local	global	Vaihingen		Potsdam	
		mAcc	mIoU	mAcc	mIoU
√		87.71	79.84	84.21	74.33
	√	87.97	80.34	84.09	74.28
√	√	**89.22**	**81.27**	**84.70**	**75.24**

TABLE VI
ABLATION EXPERIMENTS USING A W/O APPROACH

	Vaihingen		Potsdam	
	mAcc	mIoU	mAcc	mIoU
W/O WAM in local	88.46	80.70	84.31	74.39
W/O WAM in global	88.55	80.41	84.45	74.45
W/O WAM in both	87.15	79.36	84.06	74.34
W/O LSCU	88.67	80.61	84.50	74.86
Ours	**89.22**	**81.27**	**84.70**	**75.24**



(a) (b) (c) (d) (e) (f)

Fig. 5. Dense prediction comparison (a) IRRG, (b) Label, (c) Baseline, (d) Baseline + CIM, (e) Baseline + WCHM, and (f) HLMamba.

TABLE VII
GENERALIZATION ABILITY VERIFICATION USING RGBD-MIRROR

Method	IoU↑	F_β↑	BER↓	MAE↓
MirrorNet$_{2021}$ [65]	68.37	0.723	8.66	0.062
SANet$_{2022}$[66]	78.43	0.887	8.16	0.041
VCNet$_{2023}$ [67]	73.01	0.848	10.42	0.052
SATNet$_{2023}$ [68]	78.42	0.906	9.31	0.031
CSFwinformer-B$_{2024}$ [69]	78.66	0.900	8.57	0.031
MedMamba$_{2021}$[50]	78.37	0.812	9.70	0.048
HLMamba	78.90	0.890	9.44	0.041

2) Contribution of the WCHM: Tables IV and V confirm the overall usefulness of the WCHM. In particular, WCHM is mainly divided into a global region comprising the 2DSSM and a local region comprising the LSCU, which aggregates global and local multiscale features delivered by the CIM, respectively. By adding WCHM on top of the baseline, the model metrics increased for both datasets. In contrast, subtracting it from the full LHMamba reduces the performance (Fig. 5).

To explore whether the designed WCHM adequately combines local and global features, we designed experiments that included only the local or global parts. Table V shows the obtained results, which indicate that when the WCHM has only local or global parts, the performance of the model on both datasets decreases relative to that of the full structure. Moreover, Fig. 6 shows that combining CNN and mamba in WCHM is instrumental in generating the final prediction.

3) Contribution of WAM and LSCU: The original intention in designing WAM and LSCU was to use wavelets to provide additional wavelet signals for feature aggregation and capture local feature relationships using basic convolution in

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TABLE VIII
PERFORMANCE COMPARISON BASED ON LOVEDA

	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU
U-Net$_{2015}$[52]	40.13	53.48	57.10	64.12	29.30	40.26	42.65	46.72
SegNet$_{2017}$[4]	42.68	65.38	61.71	72.61	28.30	45.07	51.25	52.42
HRCNet$_{2021}$[5]	41.41	64.07	60.48	74.32	24.78	42.04	57.81	52.13
MANet$_{2022}$[6]	44.38	56.2	56.03	78.41	17.06	45.91	58.17	50.87
CMX$_{2022}$[22]	40.59	64.19	60.54	74.19	27.31	40.67	56.37	51.98
FTUNetFormer$_{2022}$[10]	38.07	61.47	60.36	66.48	36.51	38.76	42.5	49.16
ShelfNet[71]	41.94	61.67	57.48	68.37	14.21	42.56	47.20	47.63
DC-Swin[35]	42.30	63.89	64.32	72.00	33.46	43.68	48.05	52.53
MedMamba$_{2024}$[50]	43.50	60.17	58.44	69.41	31.28	39.17	44.11	49.44
Ours	46.03	62.89	59.03	77.91	25.07	44.01	58.13	53.01

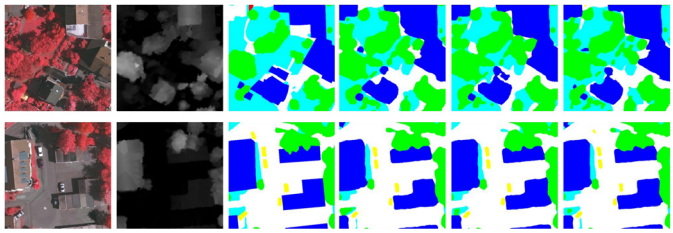


Fig. 6. Dense prediction comparison (a) IRRG, (b) nDSM, (c) Label, (d) with full WCHM, (e) only with local, and (f) only with global.

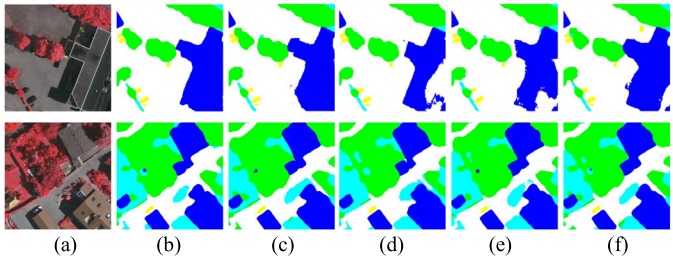


Fig. 7. Dense prediction comparison (a) IRRG, (b) HLMamba, (c) without WAM in local, (d) without WAM in global, (e) without WAM in both, and (f) without LSCU.

CNNs. The data in Table VI indicate that the performance of LHMamba is reduced by the absence of any of these modules (Fig. 7), where the absence of these modules results in errors in the prediction of some pixels on the boundary. Fig. 7 illustrates the effectiveness of the proposed WAM and LSCU.

E. Generalization Ability Verification

Two additional datasets were used to validate the generalization ability of the model. Moreover, instead of the RSIs dataset, the mirror dense prediction dataset RGBD-Mirror [65], which uses dual-stream normal RGB images and depth as inputs, was used. This dataset contained 3049 pairs of RGB-D images, of which 2000 were used for training and 1049 for testing. We compared HLMamba and MedMamba [50] with five SOTA methods for mirror dense prediction tasks: MirrorNet [65], SANet [66], VCNet [67], SATNet [68], and CSFwinformer-B [69]. In addition, four evaluation metrics from the mirror dense prediction: IoU, F-measure (F_β), balance error rate (BER), and mean absolute error (MAE), were computed. The results are summarized in Table VII. The results show that mamba-based models are promising for the mirror dense prediction. Among them, HLMamba improves existing mamba structure. Hence, the proposed solution is advantageous and shows good generalization ability on other dense prediction datasets.

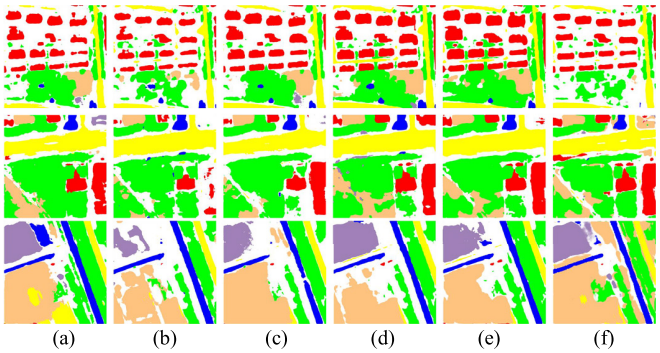


Fig. 8. Dense prediction comparison (a) HLMamba, (b) MedMamba, (c) ShelfNet, (d) dc-Swin, (e) CMX, and (f) HRCNet.

Furthermore, we used the LoveDA urban dataset [70], which contains RSIs dataset captured from three cities in China: Wuhan, Nanjing, and Changzhou, comprising 1833 images at a resolution of 1024 × 1024 pixels. In contrast to the previous two datasets, this dataset included seven predefined categories: background, building, road, water, barren land, forest, and agriculture. The experimental setup was identical to the previous setup, except for the resolution. We used 1156 of these as the training set and the remaining 677 as the test set. We compared HLMamba and MedMamba [50] with eight SOTA methods, including U-Net [54], SegNet [4], HRCNet [5], MANet [6], CMX [22], FTUNetFormer [10], ShelfNet [71], and dc-Swin [35]. Moreover, we employed two evaluation metrics: IoU and mIoU. The results are listed in Table VIII, demonstrating that HLMamba achieved good performance on the LoveDA urban dataset, particularly in the classification of water, agriculture, and background. Simultaneously, Fig. 8 shows a visualization of the dense predictions of several models. The figure shows that our model performed well, demonstrating excellent generalization ability on other RSI datasets.

F. Studies Under Extreme Environmental Settings

We experimented with extreme environmental settings using the Vaihingen and Potsdam datasets to show the applicability of the HLMamba. Specifically, we followed Cui et al. [72] to create three degradation settings: incorporating random blur,

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TABLE IX
STUDIES UNDER THREE EXTREME ENVIRONMENTAL SETTINGS

	Vaihingen		Potsdam	
	mAcc	mIoU	mAcc	mIoU
+Random Blur	88.47	80.50	84.25	74.47
+Random Noise	87.21	79.05	83.26	74.01
+both	86.22	78.29	82.41	73.28
Ours	**89.22**	**81.27**	**84.70**	**75.24**

random noise, and both of them, where the degradation factor was set to two. The results are listed in Table IX.

In addition, although the proposed model shows a decline in performance under three different extreme settings, the reduction is relatively minor, demonstrating its applicability to real-world environments. These results suggest that combining global and local features is effective in mitigating the negative effects of degradation.

V. CONCLUSION

In this study, we introduced HLMamba, a lightweight hybrid mamba fusion model for dense prediction tasks. HLMamba uses CIM and WCHM modules to achieve high performance on RSIs while maintaining a low computational cost. First, we designed CIM, which establishes connections between intra- and inter-modal features within the mamba structure to facilitate effective feature fusion. Second, we proposed WCHM, which incorporates LSCU and 2DSSM to aggregate local and global features. The parallel WAM assigns wavelet features that provide additional valuable information to the fused features. Extensive experiments conducted on the ISPRS Vaihingen and Potsdam datasets demonstrated that HLMamba outperformed the SOTA CNN-, transformer-, and mamba-based models in terms of efficiency and accuracy.

Although HLMamba outperformed in dense prediction tasks, certain limitations remain. First, the models for dense prediction tasks are relatively simple and do not fully realize the potential of the SSM. Meanwhile, the dense prediction of RSIs requires a large amount of training data, thus limiting the applicability of HLMamba for labeling incomplete data.

In future work, we aim to apply HLMamba to other RSI tasks and investigate the potential of mamba-based models in self-supervised learning to stimulate their further development. We will further improve HLMamba to fully and comprehensively consider real-world scenarios and contribute to urban planning, environmental monitoring, and other fields.

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